

Trading Options and CDS on Stocks under the Short Sale Ban

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Abstract

We examine the interaction between price discovery in banned stocks and the trading and prices of options and CDS during the 2008 short sale ban. We find that among banned stocks, stocks with high open purchased put-call ratios, low synthetic to stock prices ratios, or high CDS percentage change underperform in the following five days. We also find options prices are more efficient for un-banned stocks during the ban period. Our results suggest that informed investors trade derivatives in severely deteriorated market conditions, and that derivative prices are more informationally efficient than stock prices during the ban.

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1 Introduction

On September 19, 2008, shortly after the bankruptcy filing of Lehman Brothers and the government bailout of AIG, the Securities and Exchange Commission (SEC) issued an emergency ban on short sale for all financial stocks. The original motivation for SEC to institute such a ban was to prevent further price disruptions and to manage the wide-spread crisis of confidence in financial institutions; however, its effectiveness was questionable and the unintended consequences were many.¹ For example, Battalio and Schultz (2011) examine how the confusion and regulatory uncertainty generated by the imposition of the short sale ban (SSB) affect equity options markets, and find that the short sale ban is associated with dramatically increased bid-ask spreads for options on banned stocks. Grundy, Lim, and Verwijmeren (2012) reach the same conclusion on bid-ask spreads and find a significant diminution in options volume for banned stocks relative to unbanned stocks during the ban. They conclude that the ban acted as an effective restriction on trading in options.

The main objective of this study is to examine under the severely deteriorated market condition, whether informed investors use options to trade on negative information and how derivative prices react differently to those trades during the ban. We first examine the trading behavior in the options market during the SSB. One could think of buying a put on the SSB stock as an indirect but open channel through which any potential negative information about the underlying stock could get in. Our analysis focuses on the banned financial stocks with exchange traded options over the 14-day period between September 19 and October 8.² Using data from the Chicago Board of Options Exchange (CBOE) and

¹See, for example, the New York Times opinion piece by Arturo Bris on September 29, 2008.

²Over this time period, there were additions and removals from the original list; we focus on stocks that

International Securities Exchange, we construct put call ratios from option volumes initiated by non-market maker public buyers to open new option positions. For each of the 14 days, we sort the SSB stocks by their put-call ratios into three groups: the top and the bottom 20% and the middle 60%. The stocks in the top 20% group are the ones for which options investors express the most negative information. They are also the most useful stocks for us to study, as the SSB has the potential to impede the incorporation of negative information into stock prices. We find that during the ban, this top 20% group under-performs the middle 60% group by 4.46% over the next five-day excess return. The differences are not only economically large, but they are also statistically significant, even though this test was done over a 14-day window. In contrast, the performance of the bottom 20% group, which relatively speaking options investors are relatively positive about, is not economically or statistically different from that of the middle 60% group.

By performing our analysis on a group of stocks affected by the same short sale ban and examining the predictability cross-sectionally within this group, we provide evidence that investors buy puts to trade on negative information and some of the banned stocks are over-valued during the ban. Matching each banned stock with an unbanned stock that also has exchange traded options, we perform the same analysis on the control stocks over the same ban period and find no predictive result. We also perform the same analysis over the 14-trade-day periods immediately before and after the ban, and find no result of predictability.³ These results suggest the severely deteriorated market conditions mainly eliminate traders without valuable information.

were banned throughout the ban.

³Put call ratios have symmetric predictability for stock prices during normal time period from January to July 2008.

The second objective of our study is to examine whether options prices are more informative about future stock prices during the ban. The information signal we use is the ratio of daily synthetic to actual stock price (SS). If options prices contain information of stocks, SS will be positively associated with subsequent stock returns. Indeed we find that during the ban, SSB stocks with low (high) SS underperform (outperform) the middle group by 4.55% (1.97%) in following five days. Even though SS can forecast stock returns in regular time, it has no predictability during the 14-day periods right before or after the ban. These results are consistent with that informed traders leave traces of information in the options prices for banned stocks during the ban. Additionally, we find control stocks with high SS also outperform during the ban, suggesting SSB has spillover effects on the information content of option prices of non-banned stocks.

The third result of our study focuses on the interaction between credit default swaps and stocks during the short sale ban. Given that CDS is an insurance on the potential default of a firm, it is naturally more sensitive to negative information. However, it is difficult to obtain trading information on CDS as it is traded over the counter, and our CDS sample is markedly smaller than the option sample: there are only 60 banned stocks with active CDS quotes. Nevertheless, our result shows that, in the presence of the SSB, it takes time for the negative information already incorporated in CDS spreads to affect stock prices.

For each banned stock, we construct its daily percentage change in CDS spreads, and use this as a CDS signal. Cross-sectionally, a firm with a more positive change in CDS is the one with more negative information. Connecting this signal to stocks, we find that over the 14-day short ban period, if the CDS signal of one banned stock is one standard deviation higher than that of another banned stock, this stock on average underperforms the other by 1.29%

over the next two-day return, and by 2.69% over the next five-day return. Moreover, this predictability is asymmetric, driven mostly by stocks with more positive percentage changes in CDS. Performing the same analysis over the periods immediately before and after the ban or on 60 unbanned control stocks during the ban, we do not find any predictability in percentage changes in CDS.

Conceptually, our study follows the theoretical work of Diamond and Verrecchia (1987), Miller (1977), and Chen, Hong, and Stein (2002), who state that stock prices tend to be upward-biased by short-sale prohibition or restrictions. Diamond and Verrecchia (1987) predict that when there is a short prohibition that eliminates both informed and uninformed investors, stock price will incorporate negative information with a delay. It is important to point out that the information used to gauge the speed of adjustment is publicly available: both the put-call ratio and the CDS prices can be obtained in real time for a fee.

The effect of the 2008 short ban on the options market has been studied in several recent papers. Battalio and Schultz (2011) and Grundy, Lim, and Verwijmeren (2012) show deteriorated option market liquidity and a significant decrease in option volumes. Lin and Lu (2016) argue when demand in the equity lending market is driven solely by the hedging needs of options market makers, there will be a reduction in put volume. Our study focuses on the relative information efficiency of stock and option markets during the ban and shows that informed traders still bring information to the options market when liquidity is severely deteriorated. Our result of enhanced information efficiency of options trading and prices is complement to Battalio and Schultz (2011) and Grundy, Lim, and Verwijmeren (2012). Together with their findings, our result suggests that deteriorated liquidity in the options market eliminates more noisy than informed traders, the trading and prices of options con-

sequently become more informationally efficient.

For the stock market, Boehmer, Jones, and Zhang (2013) document that the banned stocks also suffer a severe degradation in quality, as measured by spreads, price impacts, and intraday volatility. Kolasinski, Reed, and Thornock (2013) find that the price impact of short sales increased during the ban for affected stocks. Gagnon and Witmer (2009) document price differences between the U.S. and Canadian shares and trading volume migration from U.S to Canada among banking stocks during the ban. Harris, Namvar, and Phillips (2009) use a factor-analytic model to show that the ban led to substantial price inflation in the SSB stocks.

Our study is also related to Sorescu (2000), Danielsen and Sorescu (2001), and Mayhew and Mihov (2005) who focus on option introductions and investigate whether options help to relax binding short sale constraints and whether options prices contain negative information. Ofek, Richardson, and Whitelaw (2004) find that violations of put-call parity are asymmetric in the direction of short sale constraints, and the magnitudes are strongly related to the cost of short sale. Given that option introductions and the cost of short sale are generally endogenous, it is difficult to conclusively determine how the presence of the options market or the cost of short sale might affect the underlying stock market. Our study takes advantage of an event that was also highly endogenous, but imposed on a broad enough set of stocks for us to examine cross-sectional differences among the SSB stocks.

Our study is also related to the literature examining the cross-market information transmission between stocks and options. Among others, Easley, O'Hara, and Srinivas (1998) provide a formal framework to examine this issue and Pan and Poteshman (2006) present strong evidence that options trading volume contains information about future stock prices.

One important ingredient of their result is that the options volume information used in their analysis was not publicly available during their sample period. More recently and after the publication of their paper, however, the CBOE started to release such information in real time, and the predictability has decreased markedly. Hu (2014) shows that stock orders from options market makers have power in predicting stock return. Cremers and Weinbaum (2010) and Xing, Zhang, and Zhao (2010) document that options prices also contain stock directional information. The above findings indicate that this information transmission is present during normal times and the predictability is symmetric with respect to positive and negative information. We find it is also the case during the period before the financial crisis in 2008, but focus instead on the short sale ban during the crisis, and our predictability is asymmetric with respect to the nature of information: the predictability is present mostly for negative information, but not for positive information.

For the CDS market, Longstaff, Mithal, and Neis (2005) document that there is no lead-lag relation between CDS spread and stock prices. Acharya and Johnson (2007) find a negative cross-correlation between CDS changes and future stock returns for entities with credit deterioration and high CDS levels. In more recent studies, Norden and Weber (2009) and Tang and Yan (2017) document that stock prices lead CDS spread changes in daily frequency. However, in monthly frequency, Han, Subrahmanyam, and Zhou (2017) find that the term structure of CDS is associated with future stock returns.

Finally, our study is related to studies examining the effect of short sale prohibitions and restrictions on stock prices. Bris, Goetzmann, and Zhu (2007) find evidence that prices incorporate negative information faster in countries where short sales are allowed. Chang, Cheng, and Yu (2007) analyze the price effects following the addition of stocks to the short

list in the Hong Kong market and find that short sale prohibitions tend to cause stock over-valuation. Some studies use either short interest or rebate rate as a proxy for short restriction, and document that there is a negative relation between short restriction and subsequent stock returns. For example, Asquith and Meulbroek (1996), and Aitken, Frino, McCorry, and Swan (1998) use short interest as a proxy. Jones and Lamont (2002), Geczy, Musto, and Reed (2002), and D'Avolio (2002) examine the predictability of rebate rate. And Saffi and Sigurdsson (2011) document that a low stock loan supply is associated with price inefficiency.

The rest of this paper is organized as follows. Section 2 provides some details on the short sale ban and the option and CDS data used in our analysis. Section 3 presents our main result with respect to option trading, option price and future stock returns, and Section 4 presents the result relating CDS to future stock returns. Section 5 concludes the paper.

2 Short Sale Ban, Data and Summary Statistics

2.1 Short Sale Ban

On September 19, 2008, the SEC issued a short sale ban (SSB) on financial stocks. Initially, the ban covered 797 financial stocks selected by the SEC. On the following trading day, September 22, the SEC allowed exchanges to determine which firms would be included or excluded from the ban. Eventually, around 990 stocks were banned, with some stocks entering and exiting the ban list.

The ban was set to expire in 10 days, but could be extended to 30 days at the SEC's

discretion. On October 3, immediately after the rescue plan passed the Congress, the SEC announced that the ban would expire at 11:59pm ET October 8. The ban lasted 14 trading days, from September 19 to October 8. We choose August 19 to September 18 as the pre-ban period, and October 9 to November 8 as the post-ban period for this study.

2.2 Data

The data used to compute the main option trading activity are obtained from CBOE and ISE. The dataset contains daily non-market maker public customer volume for all CBOE and ISE listed options. For each option, the daily trading volume is divided into four types of trades: open buy, in which non-market makers buy options to open new positions; close buy, in which non-market makers buy options to close existing written option positions; open sell, in which non-market makers sell options to open new short positions; and close sell, in which non-market makers sell options to close out existing long options positions. Among these four types of option trades, the information content of the open buy volume is perhaps the highest, as it does not involve any pre-existing option positions or margin requirements. For this reason, our main analysis focuses on option volumes associated with open buys made by public customers. In our analysis, we also use general option volumes obtained from Option Metrics.

During the SSB period, there are 229 banned stocks and 2036 non-banned stocks with options traded on CBOE and ISE. For comparison, we match each optionable SSB stock with an optionable control stock, which is selected by minimizing the sum of the squared percentage differences of stock market capitalization, stock volume, and stock price. The

matching is same as that used in Battalio and Schultz (2011), and is done for the August 19 to November 8 period.

Options intraday price, stock bid ask price and stock short interest data are obtained from CBOE OPRA, TAQ and Markit, respectively.

2.3 Option Trading and Price Signal

The option trading signal we use is the put call ratio (PC) calculated as follows:

$$PC_t^i = P_t^i / (P_t^i + C_t^i), \quad (1)$$

where P_t^i is the number of non-market maker public investor open purchased put contracts for stock i on date t , and C_t^i is the same number for call contracts. We select sample stocks with at least 50 contracts of open purchased volume to exclude the possibility that the result is driven by stocks with little option trading.

If investors are bearish about stock they will open buy more put options and PC will increase. Pan and Poteshman (2006) shows it has strong negative predictability for subsequent stock returns. Investors can also short naked calls when they have negative information on stocks. However, Lakonishok, Lee, Pearson, and Poteshman (2006) show the most popular option strategy among retail investors is shorting covered call (buying stock and shorting call) whose payoff increases with stock prices. Since we are not able to tell which short call positions are combined with purchased stocks, we do not include short position in the option trading analysis.

To examine how the ban influences the stock directional information in options prices,

we use the ratios of synthetic stock price to stock price (SS) as the signal of stock price information in options prices. We calculate this as follows:

$$SS_t^i = \frac{C_t^i + K^{-r(T-t)} - P_t^i + EEP_t^i}{S_t^i} \quad (2)$$

In the above equation, C_t^i and P_t^i are the prices of ATM calls and puts expiring next month at 4:00pm close; S is the stock close price at the same time. EEP is the early exercise premium of American put, and computed as the price difference between binominal American and Black-Scholes models with volatility estimated from past 30 trading day returns.⁴ Ofek, Richardson, and Whitelaw (2004), Cremers and Weinbaum (2010) and Xing, Zhang, and Zhao (2010) find SS or a monotonic transform of SS predict stock returns positively.

Table 1 summarizes the option trading activities and prices of the SSB and control stocks before, during, and after the short sale ban. For each variable, we compute the cross-sectional mean, median and standard deviation for every trading day and report their time series averages. In our main analysis, we use the put-call ratio (PC) and synthetic to stock (SS) as information signals in the options market. There is a decreased activity in open bought put relative to call contracts for the SSB stocks during the ban. This can be seen in two ways. First, the mean and median of the PC of the SSB stocks are lower during the ban than before and after the ban. Second, the difference between the PC of the SSB and the control stocks is also lower during the ban. This result is consistent with the finding of Grundy, Lim, and Verwijmeren (2012), who show that put volume of the SSB stocks decreases during the ban.

⁴We delete put call pairs with stock dividend payment and strike prices below 90% or larger than 110% of stock price.

Table 1: Summary Statistics of Stocks with Options

	SSB Stocks			Control Stocks		
	mean	median	std	mean	median	std
Panel A: Ban Period, Sep 19 - Oct 8						
Put Volume (contracts)	713,464	71,368	2,299,883	607,386	60,804	1,955,880
Call Volume (contracts)	779,988	66,850	2,602,779	642,717	76,757	1,944,299
Option/Stock Volume	0.128	0.069	0.204	0.122	0.062	0.200
Put/Stock Volume	0.068	0.031	0.138	0.058	0.023	0.128
Call/Stock Volume	0.060	0.033	0.097	0.064	0.032	0.103
Open Buy p/p+c (PC)	0.471	0.451	0.369	0.435	0.390	0.366
SyntheticS/StockS (SS)	0.998	1.000	0.025	1.000	1.000	0.022
Option Bid Ask Spread (BO)	0.337	0.281	0.225	0.257	0.206	0.176
Panel B: Before Ban Aug 29 - Sep18						
Put Volume (contracts)	972,510	84,708	3,372,184	658,505	62,092	2,009,665
Call Volume (contracts)	809,980	87,708	3,015,430	712,102	73,912	2,186,903
Option/Stock Volume	0.112	0.058	0.159	0.132	0.063	0.220
Put/Stock Volume	0.058	0.024	0.090	0.058	0.023	0.105
Call/Stock Volume	0.054	0.027	0.089	0.074	0.032	0.144
Open Buy p/p+c (PC)	0.500	0.506	0.362	0.417	0.362	0.367
Synthetic/Stock (SS)	0.997	1.000	0.012	1.000	1.001	0.009
Option Bid Ask Spread (BO)	0.233	0.194	0.155	0.206	0.176	0.135
Panel C: After Ban Oct 9 - Oct28						
Put Volume	733,014	76,193	2,201,612	668,849	69,368	2,072,386
Call Volume	658,743	87,571	1,904,772	606,150	83,729	1,898,124
Option/Stock Volume	0.107	0.054	0.160	0.108	0.057	0.164
Put/Stock Volume	0.059	0.022	0.113	0.052	0.022	0.098
Call/Stock Volume	0.048	0.027	0.068	0.056	0.029	0.093
Open Buy p/p+c (PC)	0.478	0.455	0.365	0.421	0.372	0.360
Synthetic/Stock (SS)	0.992	0.996	0.026	0.998	1.00	0.017
Option Bid Ask Spread (BO)	0.297	0.249	0.190	0.265	0.218	0.172

Reported are the time-series averages of daily cross-sectional mean, median, and standard deviation. Option/Stock Volume (OS) is 100 times the number of option contracts divided by stock volume. Call/Stock Volume and Put/Stock Volume are similar variables for call and put contracts. P/(P+C) Open Buy Volume is the number of put open buy contracts from public investors divided by put plus call open buy contracts from public investors. BO is the volume weighted option price bid ask spread in dollars.

The summary statistics of the general option volume also suggest that trading in the option market do not increase during the ban. The option volume of the SSB stocks during the ban is lower than before the ban; the decrease is more obvious for puts than calls. Though the option/stock volume of the SSB stocks is higher during the ban, the enhanced relative option volume should come from decreased stock volume, as the call and put volumes do not increase during the ban.

2.4 CDS Data

The CDS data used in this paper are from CMA via Datastream. During the short ban period, there are 395 stocks with CDS daily closing quotes, among which 60 are SSB stocks. We select 60 control stocks from the non-SSB stocks using the same matching method as before. Specifically, control stocks are selected by matching stock market capitalization, stock trading volume and stock price.

Table 2 summarizes the CDS data during, before and after the ban. The SSB stocks on average have higher CDS spread than the control stocks during the ban, and this is reflected in both the cross-sectional mean and median. For example, the average CDS spread is 629.63 bps for the SSB stocks, compared with 452.91 bps for the control stocks. Before and after the ban, however, this relation is somewhat mixed, depending on whether the mean or median is used in the comparison. We also find that the SSB sample is more dispersed with a higher cross-sectional standard deviation. Overall, there is a general pattern of increasing CDS spreads for both the SSB and the control stocks as the broad economic condition worsens over the sample period.

Table 2: **Summary Statistics of Stocks with CDS**

	SSB			Control		
	mean	median	std	mean	median	std
Panel A: Ban Period, Sep 19 - Oct 8						
CDS (%)	5.58	1.19	8.73	3.85	1.47	5.00
dCDS (%)	0.20	0.01	1.13	0.12	0.02	0.48
CBA (%)	0.91	0.31	1.32	0.85	0.23	1.43
Panel B: Before Ban, Aug 19 - Sep18						
CDS (%)	4.35	1.05	8.51	3.39	1.34	4.37
dCDS (%)	0.10	0.01	1.58	0.09	0.01	0.28
CBA (%)	0.64	0.21	1.13	0.67	0.19	1.21
Panel C: After Ban, Oct 9 - Nov 08						
CDS (%)	680.51	160.28	1006.36	720.57	282.26	984.40
dCDS (%)	4.52	2.25	133.85	28.22	4.38	197.96
CBA (%)	120.40	37.00	167.71	177.54	47.21	352.74

Reported are the time-series averages of daily cross-sectional mean, median, and standard deviation. CDS is the credit default spread in basis points. dCDS is daily change of CDS. CBA is the difference between CDS ask and bid.

3 Option and future stock returns

3.1 Main Results

In this section, we use option signals to investigate whether there are informed investors trade options on banned stocks during the ban. During the ban, investors in the option market can still purchase puts to bet on declining stock price. If option trading contains information, and stock prices incorporate negative information with a delay, we expect an option signal with negative information to have stronger predictability for future stock returns than an option signal with positive information. Our main empirical investigation is to test this asymmetric predictability for the SSB stocks during the SSB period. We expect this particular predictability to be less obvious for control stocks during the ban or the SSB stocks during non-ban periods.

Table 3: Returns of Open Buy P/(P+C) and Syn/S Quintiles

	Sorted on Put-Call Ratios (PC)				Sorted on Syn-Stock Ratios (SS)			
	SSB stocks		Control stocks		SSB stocks		Control stocks	
	Raw ret	Ex ret	Raw ret	Ex ret	Raw ret	Ex ret	Raw ret	Ex ret
SSB Period 2008 Sept 19 - Oct 8								
1 Low	-12.50	1.68	-11.33	2.90	-17.22	-0.35	-13.11	1.29
2	-11.44	2.81	-12.83	3.63	-13.73	1.61	-12.64	1.14
3	-12.41	0.36	-10.75	2.33	-12.57	2.91	-12.33	0.62
4	-12.90	-2.04	-11.80	-0.64	-12.52	1.95	-12.85	0.13
5 High	-15.89	-4.23	-10.93	0.58	-9.80	5.11	-9.37	3.32
3-1	0.09	-1.65	0.58	-0.57	4.64**	3.27**	0.77	-0.66
<i>t</i>	(0.11)	(-1.66)	(0.69)	(-0.67)	(5.91)	(2.68)	(0.82)	(-0.70)
5-3	-3.47**	-4.26**	-0.18	-0.74	2.77**	2.20	2.96**	2.69**
<i>t</i>	(-3.05)	(-3.52)	(-0.14)	(-0.78)	(2.65)	(1.93)	(3.00)	(2.65)
5-1	-3.39**	-5.91**	0.40	-1.31	7.41**	5.47**	3.73**	2.03**
<i>t</i>	(-2.75)	(-3.52)	(0.42)	(-1.14)	(6.08)	(4.03)	(3.47)	(2.32)
Before SSB Period 2008 Aug 19 - Sept 18								
3-1	-1.35	-1.94	0.21	0.40	-1.96**	-0.15	-4.97**	-3.39**
<i>t</i>	(-1.49)	(-1.60)	(0.28)	(0.53)	(-2.04)	(-0.25)	(-3.83)	(-2.82)
5-3	2.79**	2.78	0.72	0.42	1.27	1.94**	1.02	1.43
<i>t</i>	(2.18)	(2.42)	(0.83)	(0.45)	(1.22)	(2.19)	(1.24)	(1.90)
5-1	1.44	0.84	0.93	0.82	-0.69	-1.69**	-3.94**	-1.96
<i>t</i>	(1.68)	(0.22)	(0.87)	(0.63)	(-0.43)	(-2.06)	(-2.17)	(-1.54)
After SSB Period 2008 Oct 19 - Nov 8								
3-1	-1.83	-1.69	0.58	-0.17	-0.70	-0.62	-2.52	-2.82**
<i>t</i>	(-1.75)	(-1.71)	(0.87)	(-0.24)	(-0.60)	(-0.61)	(-1.97)	(-2.36)
5-3	-1.42	-1.14	0.56	1.01	2.10	1.01	-1.72	-1.26
<i>t</i>	(-1.46)	(-1.25)	(0.81)	(1.51)	(1.69)	(0.87)	(-1.55)	(-1.39)
5-1	-3.25**	-2.83**	1.14	0.84	1.40	0.39	-4.24**	-4.08**
<i>t</i>	(-2.89)	(-2.45)	(1.61)	(0.69)	(1.36)	(0.41)	(-3.06)	(-3.16)
Regular Period 2008 Jan 2 - Jul 31								
3-1	-0.32	-0.34	-0.48**	-0.35**	0.17	0.34**	0.08	0.11
<i>t</i>	(-1.81)	(-1.96)	(-3.80)	(-2.47)	(0.96)	(2.08)	(0.58)	(0.83)
5-3	-0.31	-0.31	-0.20	-0.32**	0.09	0.20	0.32**	0.09
<i>t</i>	(-1.91)	(-1.95)	(-1.75)	(-2.25)	(0.67)	(1.39)	(2.45)	(0.72)
5-1	-0.63**	-0.64**	-0.68**	-0.67**	0.26	0.54**	0.40**	0.20
<i>t</i>	(-2.42)	(-3.31)	(-3.96)	(-4.01)	(1.45)	(2.95)	(2.48)	(1.57)

Quintiles are formed based on open buy put to call ratios or synthetic to actual stock prices ratios on day t . Returns are in percentage and from day t to $t + 5$. Raw ret is the equal weighted average raw returns and Ex ret is the equal weighted excess return adjusted for Fama-French 3 and momentum factors. T-stats adjusted for Newey West (1987) serial correlation are in parentheses.

Table 3 reports the future five day returns of the portfolios sorted on put call ratios (PC) and synthetic to real stock prices ratios (SS). We sort stocks into quintiles for every trading day, then compute each quintile's following five trading day cumulative returns. We report equal weighted raw and risk adjusted returns. When constructing the risk-adjusted returns, we use a four-factor model of market, size, value and momentum to remove the systematic component, and we estimate the factor betas using daily returns of past 30 trading days.

Column two to five are the returns of the portfolios sorted on PC . Consistent with the argument that negative information spreads more slowly on stock prices under the SSB, we see that for the SSB stocks during the ban, the high PC quintile has the lowest future returns, and the differences between the high and the third quintiles are all significantly negative, whereas the differences between the third and the first quintile are not statistically different from zero. For example, in five day raw returns, the high PC quintile under-performs the third quintile by 3.47% (t -stats: -3.05), whereas the return difference between the third and the first quintiles is only 0.09% (t -stats: 0.11). For control stocks, the return difference between the high and the third quintiles are also negative, but not statistically significant.

During the period before the ban, PC has no predictability for both SSB and control stocks. The return differences are either not statistically different from zero or positive. During the period after the ban, PC has a little predictive power for SSB stocks but in a symmetric way, and has no predictability for the control stocks. During the normal non-crisis period, PC has predictive power for stock returns for both SSB and control stocks in a symmetric way. Most return differences between high (5) and low (1) quintiles are negative and statistically different from zero.

Column six to five are the returns of the portfolios sorted on synthetic to real stock

prices ratios (SS). If SSB stocks adjust information more slowly than their options during the ban, and options prices impacted by the informed traders, SS will be positively related to future stock returns, especially for stocks in the first (low) SS quintile. Consistent with this hypothesis, we see that for the SSB stocks during the ban, the first SS quintile has the lowest future returns, and the return differences between the third and the first quintile are all significantly positive, 4.64% (t -stats: 5.91) for the raw returns, and 3.27% (t -stats: 2.68) for the excess returns. The return differences between the fifth and the third SS quintile are also positive with statistical significance, but the magnitude is smaller than the difference with the first SS quintile, 2.77% (t -stats: 2.65) for the raw returns, and 2.22% (t -stats: 1.93) for the excess returns. For control stocks, the return differences between the third and the first SS quintile are not statistically different from zero, whereas the differences between the high and the third SS quintile are positive and some of them statistically significant.

During the period right before or after the ban, SS has no predictability for both SSB and control stocks; the return differences are either not statistically different from zero or negative. During regular non-crisis period from Jan to Jul 2008, the return differences are all positive and some of them statistically significant.

It can be seen from Table 3 that the returns are large in magnitude during the ban; the weekly raw (excess) returns are in the order of -13% (3%) for the portfolios formed during the ban. These results are due to the extremely volatile market during the ban; the average S&P500 weekly return during the same period is -9.5%.

It is possible that our options signals correlated with other factors that drive the future returns of during the ban period. To test this, we conduct a regression analysis with controls

to test the asymmetric predictability of the option signals as follows:

$$R_{t,t+\tau}^i = a + b^{PCL} PCL_t^i + b^{PCH} PCH_t^i + c BO_t^i + d R_{t-5,t}^i + f SI_t^i + g V_t^i + h BS_t^i + \epsilon_t^i \quad (3)$$

$$R_{t,t+\tau}^i = a + b^{SSL} SSL_t^i + b^{SSH} SSH_t^i + c BO_t^i + d R_{t-5,t}^i + f SI_t^i + g V_t^i + h BS_t^i + \epsilon_t^i \quad (4)$$

where $R_{t,t+\tau}^i$ with $\tau = 1, 2, \dots, 5$ is the future τ -day cumulative four factor excess return for stock i . PCL_t^i and PCH_t^i are dummies for low and high put call ratios, respectively. If stock i is in the low put call ratio (PC) quintile on date t , we assign PCL_t^i to be one, and zero otherwise. Likewise, if stock i is in the high PC quintile on date t , we assign PCH_t^i to be one, and zero otherwise. In equation 4, SSL_t^i and SSH_t^i are dummies for low and high synthetic to real stock prices ratios, respectively. For the controls, we include options volume weighted bid ask spread (BO_t^i), past five day stock return, $R_{t-5,t}^i$, short interest relative to shares outstanding, SI_t^i , stock Garch (1,1) volatility, V_t^i , and stock relative bid ask spread at market close, BS_t^i . We include them because bid ask spread, short interest, past return and volatility may be correlated with options trading and influence returns.

If short sale ban slows down the adjustment of stock prices to negative information and this negative information is contained in the trading of put options, we expect the coefficient b^{PCH} to be negative for the sample of SSB stocks during the ban. The coefficient b^{PCL} picks up the positive side of the information content and is less affected by the ban, we expect it to be less significant. Moreover, this asymmetry between b^{PCH} and b^{PCL} would mainly present for the SSB stocks during the ban, but not so for the control stocks during the ban or the SSB stocks in non-ban periods. Similarly, if synthetic stock prices from option prices

are influenced by informed trading and more informationally efficient than the real stock prices, we expect negative b^{SSL} and positive b^{SSH} . Since b^{SSL} pick up the negative side of the information, its absolute value should be higher than that of b^{SSH} during the ban.

Our results for put call ratios are summarized in the top panel of Table 4. For the sample of SSB stocks during the ban, the coefficients on PCH are negative and statistically significant from day 3 to 5, whereas the coefficients on PCL are mixed in sign and not significantly different from zero. When the dependent variable is future three-day (five-day) returns, the coefficient estimate on PCH is -2.11 with t -stats -2.00 (-4.23 with t -stats -2.97). This result indicates that relative to the average SSB stocks, those SSB stocks in the top put call ratio(PC) quintile underperform by 2.11% and 4.23% over the next three and five business days, respectively.

The right top Panel of Table 4 reports the results for the control sample during the ban. We do not find any predictability from put call ratios: neither the coefficients on PCH nor those on the PCL are statistically significant. Table 4 also shows the coefficient estimates on options bid ask spreads and lagged returns are negative. Most of those coefficient estimates are not statistically different from zero. The coefficient estimates on lagged returns are also negative.

The results for synthetic to real stock price ratios are reported in the bottom panel of Table 4. For the SSB stocks during the ban, the coefficients on SSL are negative and statistically significant from day 2 to 5. When the dependent variable is future two-day (five-day) returns, the coefficient estimator on SSL is -3.21 with t -stats -2.82 (-3.88 with t -stats -3.40). The coefficients estimates on SSH are positive and marginally statistically significant. Contrary to SSL whose coefficient estimates increases over time in magnitude,

the coefficient estimates on SSH are not much different from day-one to day-five, suggesting the main predictability only presents on day one. For the control stocks, the coefficients estimates on SSL are not consistent in signs, and the coefficient estimates on SSH are positive and become statistically significant on day four and day five. The above result indicates synthetic stock prices are more informationally efficient than real stock prices for the SSB stocks during the ban, and the SSB stocks may have a spillover effect on some of the control stocks with high synthetic to real stock price ratios.

Table 4: Future Stock Returns and Option Trading

SSB stocks during the ban								Control stocks during the ban							
τ	PCL_t^i	PCH_t^i	BO_t^i	$R_{t-5,t}^i$	SI_t^i	V_t^i	BS_t^i	PCL_t^i	PCH_t^i	BO_t^i	$R_{t-5,t}^i$	SI_t^i	V_t^i	BS_t^i	
1	0.21 (0.35)	-0.27 (-0.56)	-0.12 (-3.30)	-0.06 (-1.18)	-0.31 (-1.24)	0.17 (0.66)	4.70 (1.38)	0.50 (0.87)	0.61 (1.64)	-0.06 (-1.50)	-0.03 (-0.51)	0.20 (0.21)	0.02 (0.90)	2.74 (1.31)	
2	0.62 (1.09)	-1.68 (-1.57)	-0.14** (-2.57)	-0.08 (-1.11)	-1.23 (-1.06)	0.28 (0.92)	-0.31 (-0.08)	0.39 (0.61)	0.57 (1.12)	-0.01 (-0.23)	-0.09 (-1.33)	2.27 (0.71)	-0.18 (-0.75)	1.78 (1.02)	
3	-0.73 (-1.28)	-2.11** (-2.00)	-0.15** (-2.10)	-0.04 (-0.54)	-1.83 (-1.83)	0.25 (1.26)	-2.42 (-0.53)	0.37 (0.45)	-0.26 (-0.45)	0.00 (0.03)	-0.06 (-0.90)	2.37 (1.17)	-0.14 (-0.73)	2.08 (1.21)	
4	-1.80 (-1.48)	-3.73** (-2.29)	-0.11 (-1.21)	0.02 (0.22)	-1.46** (-2.08)	0.30 (1.46)	-1.11 (-0.21)	0.13 (0.18)	-0.41 (-0.86)	0.02 (0.48)	-0.07 (-0.84)	4.38 (1.31)	-0.15 (-0.69)	2.69 (1.19)	
5	-1.82 (-1.45)	-4.23** (-2.97)	-0.15 (-1.73)	0.09 (1.04)	-1.24* (-1.93)	0.45 (1.69)	1.61 (0.26)	-0.10 (-0.11)	-0.60 (-0.89)	0.05 (1.37)	0.00 (-0.04)	4.35 (1.82)	-0.18 (-0.51)	2.51 (1.05)	

SSB stocks during the ban								Control stocks during the ban							
τ	SSL_t^i	SSH_t^i	BO_t^i	$R_{t-5,t}^i$	SI_t^i	V_t^i	BS_t^i	SSL_t^i	SSH_t^i	BO_t^i	$R_{t-5,t}^i$	SI_t^i	V_t^i	BS_t^i	
1	-1.26 (-1.61)	1.97** (2.38)	-0.10** (-3.08)	-0.05 (-1.23)	-0.14 (-0.45)	0.15 (0.85)	4.44 (1.32)	-0.41 (-1.14)	0.83 (1.89)	-0.05 (-1.54)	-0.02 (-0.45)	-0.03 (-0.47)	0.03 (0.20)	2.22 (1.15)	
2	-3.21** (-2.82)	2.26* (1.91)	-0.09 (-1.90)	-0.05 (-0.92)	-0.25 (-0.74)	0.27 (1.12)	-1.98 (-0.05)	-0.48 (-0.65)	0.73 (0.59)	0.00 (-0.03)	-0.08 (-1.27)	-0.09 (-1.38)	-0.26 (-1.12)	1.80 (1.57)	
3	-2.93** (-2.83)	2.46* (1.96)	-0.10 (-1.46)	-0.01 (-0.15)	-0.96 (-1.53)	0.37 (1.53)	-2.15 (-0.60)	0.03 (0.05)	1.57 (1.23)	-0.01 (-0.28)	-0.05 (-0.93)	-0.06 (-1.06)	-0.10 (-1.27)	2.54 (1.46)	
4	-3.13** (-2.81)	2.28 (1.86)	-0.08 (-0.83)	0.05 (0.81)	-1.54 (-1.95)	0.29** (2.18)	-1.12 (-0.21)	-0.35 (-0.42)	2.22* (1.96)	0.00 (-0.07)	-0.06 (-0.69)	-0.08 (-0.92)	-0.13 (-1.05)	2.70 (1.26)	
5	-3.88** (-3.40)	1.82 (1.54)	-0.10 (-1.16)	0.12 (1.56)	-1.21 (-1.74)	0.12 (2.17)	2.11 (0.35)	-1.61 (-1.40)	3.24** (2.52)	0.02 (0.47)	-0.01 (-0.12)	-0.04 (-0.37)	-0.18 (-0.81)	2.80 (1.22)	

Daily cross-sectional regression of cumulative excess returns $R_{t,t+\tau}^i$ (%) on the high and the low put/call dummies (PCH_t^i and PCL_t^i), and on the high and low synthetic to real stock prices dummies (SSH_t^i and SSL_t^i). On each day t , stocks within the respective sample are sorted by open buy put/call ratios (or synthetic real stock prices ratios), those ranked above 80% have $PCH_t^i = 1$ ($SSH_t^i = 1$) and those below 20% have $PCL_t^i = 1$ ($SSL_t^i = 1$). Controls include volume weighted options bid ask spread, BO_t^i , past five day stock return, $R_{t-5,t}^i$, short interest relative to shares outstanding, SI_t^i , stock Garch(1,1) volatility, V_t^i , and stock relative bid ask spread at market close, BS_t^i . Fama-MacBeth t-stats adjusted for Newey and West (1987) serial correlations are reported in braces. Coefficient estimates on the constant are omitted.

3.2 Daily Coefficients

Given that the short sale ban lasts only for 14 trading days, and that the market is extremely volatile during the sample period, it is possible that predictability of the put call ratios and synthetic to real stock price ratios of the SSB stocks is driven by a few extreme observations. To examine this possibility, we plot the daily coefficient estimates from the Fama-Macbeth regression on the high put call ratio dummy (*PCH*) and low synthetic to stock dummy (*SSL*). The results are shown in Figure 1, where the dependent variables are future five-day returns. We can see that *PCH* and *SSL* can predict banned stock returns on most of the trading days during the ban. Twelve out of fourteen coefficients on *PCH* (or *SSL*) are negative. Thus our finding is not driven by a few extreme observations.

It is worth noting the first trading day of the ban, September 19. On that day, SEC granted option market makers sell short until 11:59 pm on the same day to facilitate the expiration of options; this suggested that on that day, option market makers anticipate they would not be able to make short sales afterward.⁵ In the early morning of the second trading day of the ban, however, the SEC amended the ban and allowed option market makers to short sell, in view of option market not function if market makers are not allowed to short.⁶ Figure 1 show that on the first day of the ban, *PCH* has a positive coefficient, while *SSL* has the highest predictive power among 14 trading days. This result implies market makers increase put price too high in the face of short sale prohibition, and buying orders from informed traders cannot be executed.

⁵See SEC Release 34-58572, September 18, 2008.

⁶See 'Options Market Makers Get Relief from SEC Ban on Short-Selling', Traders Magazine Online News, September 22,2008.

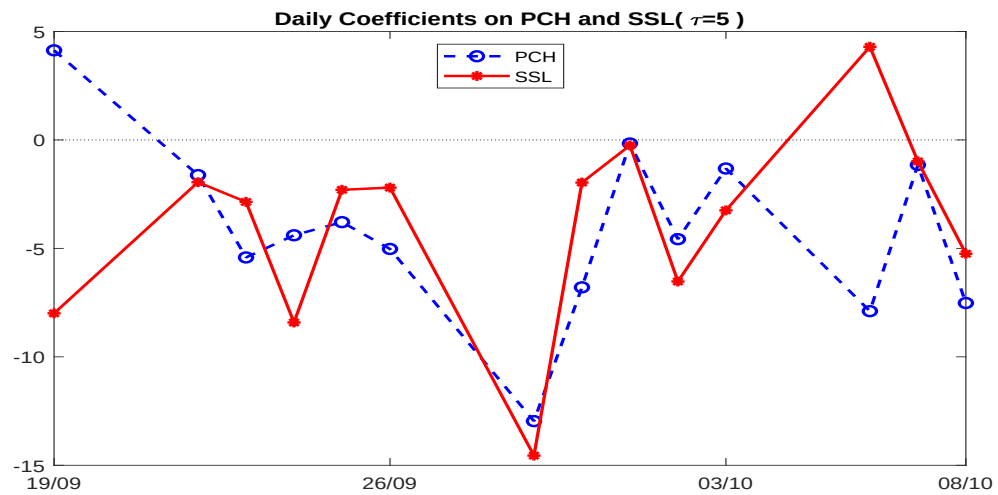


Figure 1: **Daily Coefficient Estimates on High Put Call Ratio Dummy (*PCH*), and Low Synthetic Stock Price Ratio Dummy (*SSL*)** The dependent is future 5 day cumulative excess returns. Other independent variables are low put call ratio dummy or high synthetic stock price dummy, log sizes, lagged five day returns, average implied volatility of ATM options, and average option bid ask spreads.

3.3 Spillover Effects

Boehmer, Jones, and Zhang (2018) show the total impact of a regulatory change consists of direct effects and indirect effects (spillovers). The ban may have spillover effects on the trading and informational efficiency on the non-banned stocks. To examine the potential spillover effect of the ban, we pool all observations for the periods before, during, and after the ban from August 19th to November 8th and conduct the analysis as follows:

$$R_{t,t+\tau}^i = a + b_1^{PC} PC_t^i \cdot D_t \cdot T^i + b_2^{PC} PC_t^i \cdot D_t + b_3^{PC} PC_t^i \cdot T^i + cR_{t-5,t}^i + dBO_t^i + fV_t^i + gBS_t^i + hD_t + jT_i + kD_t \cdot T_i + lSI_t^i + \epsilon_t^i \quad (5)$$

$$R_{t,t+\tau}^i = a + b_1^{SS} SS_t^i \cdot D_t \cdot T^i + b_2^{SS} SS_t^i \cdot D_t + b_3^{SS} SS_t^i \cdot T^i + cR_{t-5,t}^i + dBO_t^i + fV_t^i + gBS_t^i + hD_t + jT_i + kD_t \cdot T_i + lSI_t^i + \epsilon_t^i \quad (6)$$

In the above equations, D_t is a dummy for the SSB period and T^i is a dummy for banned stocks. The coefficients b_1^{PC} and b_1^{SS} measure the direct effect of ban on the banned stock for the informational role on PC and SS during the SSB period. As suggested by Boehmer, Jones, and Zhang (2018) the coefficients b_2^{PC} and b_2^{SS} measure the spillover effect of the ban on the non-banned stocks.

The results for PC are reported in the top panel of Table 5. The spillover effects on non-banned stocks measured by the coefficients b_2^{PC} are not consistent in signs and statistically not different from zero, suggesting the ban does not have an indirect effect on the informed option trading for non-banned stocks.

The results for SS are presented in the bottom panel of Table 5. The coefficient b_2^{SS} are all positive and statistically significant for day four and five, suggesting improved information efficiency of option prices of banned stocks positively impacts that of the non-banned stocks.

Table 5: Future Stock Returns, Option Trading and Price, Difference in Difference

Open Buy Put Call Ratios (PC)												
τ	$PC_t^i * D_t * T_i$	$PC_t^i * D_t$	$PC_t^i * T_i$	PC_t^i	$R_{t-5,t}^i$	BO_t^i	V_t^i	BS_t^i	D_t	T_i	$D_t * T_i$	SI_t^i
1	-1.12 (-1.41)	-0.89 (-1.04)	0.16 (0.20)	0.26 (0.59)	-0.04 (-0.59)	-0.09 (-0.54)	0.05 (0.66)	1.68** (2.01)	-0.35 (-0.95)	0.55 (1.10)	-0.78 (-0.80)	0.05 (0.78)
2	-2.55* (-1.94)	-0.90 (-1.03)	0.87 (0.73)	0.41 (0.75)	-0.07 (-1.03)	-0.03 (-0.13)	0.09 (1.03)	3.06** (2.95)	-0.25 (-0.54)	0.92 (1.30)	-0.96 (-0.72)	0.09 (0.98)
3	-3.43** (-2.29)	-1.10 (-1.04)	1.62 (1.31)	0.56 (1.75)	-0.11 (-1.61)	-0.15 (-0.49)	0.18 (1.55)	3.26** (3.00)	-0.36 (-0.72)	0.95 (1.06)	-1.25 (-0.94)	0.21 (1.90)
4	-4.30** (-2.45)	-1.27 (-1.25)	2.26 (1.58)	0.60 (1.63)	-0.12 (-1.72)	-0.09 (-0.25)	0.20 (1.68)	3.41** (2.65)	-0.49 (-1.09)	1.09 (1.06)	-1.54 (-1.09)	0.27** (2.10)
5	-5.45** (-2.92)	-1.56 (-1.82)	2.89 (1.84)	0.89** (2.56)	-0.14 (-1.85)	-0.16 (-0.39)	0.22 (1.90)	4.09** (2.89)	-0.67 (-1.34)	1.27 (1.19)	-1.49 (-1.02)	0.32** (2.27)
Synthetic to Real Stock Prices Ratios (SS)												
τ	$SS_t^i * D_t * T_i$	$SS_t^i * D_t$	$SS_t^i * T_i$	SS_t^i	$R_{t-5,t}^i$	BO_t^i	V_t^i	BS_t^i	D_t	T_i	$D_t * T_i$	SI_t^i
1	15.38 (1.50)	14.59 (1.30)	-15.45 (-0.85)	-1.57 (-0.52)	-0.04 (-0.89)	-0.07 (-0.38)	0.07 (0.80)	1.71** (2.02)	-24.57** (-2.35)	16.35 (0.90)	-51.32** (-2.05)	0.07 (1.01)
2	19.74* (1.95)	15.51 (1.29)	1.43 (0.05)	-2.23 (-0.33)	-0.07 (-1.29)	0.02 (0.07)	0.11 (1.18)	3.07** (2.87)	-26.59 (-1.33)	0.31 (0.01)	-50.51 (-1.17)	0.11 (1.18)
3	20.69** (2.27)	14.31 (1.52)	13.27 (0.53)	-7.42 (-1.51)	-0.11 (-1.38)	-0.09 (-0.30)	0.19 (1.69)	3.27** (2.86)	-34.28 (-1.85)	-10.77 (-0.43)	-52.49 (-1.35)	0.23** (2.05)
4	21.57** (2.25)	18.19* (1.98)	19.75 (1.13)	-12.22 (-1.52)	-0.12 (-1.88)	-0.03 (-0.08)	0.23 (1.85)	3.45** (2.57)	-52.06** (-2.00)	-36.65 (-1.06)	-15.07 (-0.33)	0.29** (2.25)
5	23.59** (2.55)	25.62** (2.75)	16.55 (0.89)	-16.83** (-2.04)	-0.14 (-2.21)	-0.10 (-0.24)	0.25** (2.08)	4.15** (2.79)	-64.43** (-2.52)	-26.85 (-0.79)	-27.85 (-0.66)	0.33** (2.40)

Pooled regression of cumulative excess returns $R_{t,t+\tau}^i$ (in basis points) on put call ratios (PC) and synthetic to real stock prices (SS). D_t is a dummy for the SSB period and T^i is a dummy for banned stocks. Controls include volume weighted options relative bid ask spread (BO_t^i), past five day stock return, $R_{t-5,t}^i$, short interest relative to shares outstanding, SI_t^i , stock Garch(1,1) volatility, V_t^i , and stock relative bid ask spread at market close, BS_t^i . Coefficient estimates on the constant are omitted. T-stats adjusted for serial and cross correlations are reported in braces. Time period includes before, during and after ban from August 19th to November 8th 2008.

4 CDS and future stock returns

During the ban, investors in CDS market can still buy CDS to profit on negative information. Given that CDS is an insurance against the default of a firm, it is naturally a more sensitive to negative information, making it a good candidate for our analysis. Cross-sectionally, a firm with more positive change in CDS has more negative information. If the short sale ban reduces the speed of the stock price adjustment to negative information, and if trading affects CDS spread as documented in Tang and Yan (2017), we expect an increase in CDS to predict a low future stock return. To test this predictability, we use the following model:

$$R_{t,t+\tau}^i = a + b \Delta CDS_t^i + c R_{t-5,t}^i + d SI_t^i + f V_t^i + g BS_t^i + \epsilon_t^i \quad (7)$$

$$\text{where } \Delta CDS_t^i = 100\% \times (CDS_t^i - CDS_{t-1}^i) / CDS_{t-1}^i, \quad (8)$$

where $R_{t,t+\tau}^i$ is the cumulative return of stock i from trade day t to $t + \tau$, ΔCDS_t^i is the percentage change of CDS from date $t-1$ to date t . If the short sale ban slows down the stock price adjustment to negative information, and informed traders trade CDS, the coefficient on ΔCDS_t^i will be negative for the SSB stocks during the ban, but less so for the control stocks during the ban or the SSB stocks during the non-ban periods.

The result presented in Table 6 shows that the coefficient estimates on ΔCDS are negative and statistically significant only for the SSB stocks during the ban. When the dependent variable is two (five) day future returns, the coefficient estimate is -0.11 (-0.24), implying that a one standard deviation increase in ΔCDS is associated with a 1.29% (2.69%) return decrease in the next two (five) days. For the control stocks, the coefficient estimates on

Table 6: Future Stock Returns and Percentage Changes of CDS

Panel A. The SSB and control stocks during the ban										
τ	SSB stocks					Control stocks				
	ΔCDS_t^i	$R_{t-5,t}^i$	SI_t^i	V_t^i	BS_t^i	ΔCDS_t^i	$R_{t-5,t}^i$	SI_t^i	V_t^i	BS_t^i
1	-8.06 (-1.78)	-0.08 (-1.66)	-0.13 (-0.34)	0.15 (0.68)	1.78 (1.28)	0.18 (0.04)	0.01 (0.19)	-0.17 (-0.89)	-0.57** (-2.11)	-1.84 (-1.74)
2	-12.48** (-2.09)	0.01 (0.15)	-0.73 (-1.85)	-0.14 (-0.32)	1.60 (1.46)	-1.67 (-0.29)	0.02 (0.25)	-0.32 (-1.35)	-0.54 (-1.75)	-1.10 (-1.00)
3	-18.02** (-2.29)	0.00 (-0.03)	-0.99** (-2.32)	0.01 (0.02)	2.81 (1.64)	-1.50 (-0.22)	-0.03 (-0.39)	-0.39 (-1.32)	-0.77 (-3.16)	-2.28 (-1.47)
4	-16.40 (-1.84)	0.04 (0.53)	-0.91 (-1.84)	0.16 (0.29)	4.89** (2.23)	-6.25 (-1.12)	(-0.01)	-0.20 (-0.75)	-0.67** (-2.49)	-1.99 (-1.20)
5	-25.56** (-2.16)	0.13 (1.45)	-0.62 (-1.19)	0.15 (0.25)	5.30** (2.18)	-9.15 (-1.31)	0.04 (0.36)	0.01 (0.03)	-0.19 (-0.46)	-1.02 (-0.40)
Panel B. The SSB stocks during non-ban periods										
τ	Before Ban		After Ban		Regular Time Jan-Jul 2008					
	ΔCDS_t^i		ΔCDS_t^i		ΔCDS_t^i					
1	-4.90 (-1.56)		-0.97 (-0.87)		-1.57 (-1.36)					
2	-8.96 (-1.54)		4.51 (0.99)		-1.97 (-1.31)					
3	-5.89 (-0.65)		6.81 (1.18)		-1.28 (-0.42)					
4	-10.27 (-0.91)		6.03 (0.92)		-1.70 (-0.19)					
5	-8.90 (-0.75)		4.61 (0.99)		-1.27 (-0.76)					

Daily cross-sectional regression of cumulative returns $R_{t,t+\tau}^i$ on daily CDS percentage change (ΔCDS_t^i). Returns are in percentage. Controls include past five day stock return ($R_{t-5,t}^i$), short interest relative to shares outstanding (SI_t^i), stock Garch(1,1) volatility (V_t^i), and stock relative bid ask spread at market close (BS_t^i). Coefficients on the constant, and Panel B coefficient estimates on controls are omitted. Fama-MacBeth t-stats adjusted for Newey and West (1987) serial correlations are reported in braces. **: significant at 5% level.

ΔCDS are not significantly different from zero. Panel B of Table 6 reports the coefficient estimates on ΔCDS for the SSB stocks during the non-ban periods. To save space we omit the estimates on the constant and control variables. As shown in Panel B, the CDS has no predictability during the non-ban periods. Right before the ban and during the normal time period from January to July 2008, the coefficients on ΔCDS are negative but not statistically significant, and during the post ban period, the coefficients are positive and not significant.

5 Conclusions

In this study, we focus on the short sale ban of 2008 and examine the interaction between price discovery in banned stocks and the trading of options and CDS during the 14-day period. We find that investors still use derivatives to trade on negative information when the derivative market conditions are severely deteriorated. Our results also suggest that in the presence of a short sale ban it takes time for the negative information contained in either the options market or the CDS market to be incorporated into stock prices.

Given the volatile nature of the markets during the short sale ban, it is important for us to minimize the influence of the overall market movement on our predictive results. For this reason, we structure our test to focus within the group of stocks with the same short sale ban and examine the predictability cross-sectionally within this group. In other words, our finding of sluggishness in price impoundment is a measure of one banned stock relative to another banned stock. The only thing differentiating the paired stocks is that one has some negative information according to the options or CDS markets.

In extracting the information from the options market, we first use the trading information. Given the volatile nature of the time period and the large bid/ask spreads in the options market, this action variable could perhaps cleanly capture option information. We then use options prices to show that synthetic stock prices are also more informative for future stock prices during the SSB. This result suggests that increased bid ask spreads reduce number of noise traders to the extent that options prices become more informational efficient. The results on CDS confirm that in the presence of short sale ban and large transaction costs, it takes time for the information already incorporated in CDS prices to be incorporated into stock prices.

References

- Acharya, V. V. and T. C. Johnson (2007). Insider trading in credit derivatives. *Journal of Financial Economics* 84, 110–141.
- Aitken, M. J., A. Frino, M. S. McCorry, and P. L. Swan (1998). Short Sales are Almost Instantaneously Bad News: Evidence from the Australian Stock Exchange. *Journal of Finance* 53, 2205–2223.
- Asquith, P. and L. Meulbroek (1996). An Empirical Investigation of Short Interest. Working Paper, Harvard University.
- Battalio, R. and P. Schultz (2011). Regulatory Uncertainty and Market Liquidity: The 2008 Short Sale Ban’s Impact on Equity Option Markets. *The Journal of Finance* 66(6), 2013–2053.
- Boehmer, E., C. M. Jones, and X. Zhang (2013). Shackling Short Sellers: The 2008 Shorting Ban. *The Review of Financial Studies* 26, 1363–1400.
- Boehmer, E., C. M. Jones, and X. Zhang (2018). Potential Pilot Problems: Treatment Spillovers in Financial Regulatory Experiments. Working Paper, Singapore Management University - Lee Kong Chian School of Business, Columbia Business School, Tsinghua University - PBC School of Finance.
- Bris, A., W. N. Goetzmann, and N. Zhu (2007). Efficiency and the Bear: Short Sales and Markets around the World. *Journal of Finance* 62, 1029–1079.
- Chang, E. C., J. W. Cheng, and Y. Yu (2007). Short-Sales Constraints and Price Discovery: Evidence from the Hong Kong Market. *Journal of Finance* 62, 2097–2121.
- Chen, J., H. Hong, and J. C. Stein (2002). Breadth of Ownership and Stock Returns. *Journal of Financial Economics* 66, 171–205.
- Cremers, M. and D. Weinbaum (2010). Deviations from Put-Call Parity and Stock Return Predictability. *Journal of Financial and Quantitative Analysis* 45, 335–367.
- Danielsen, B. and S. Sorescu (2001). Why Do Option Introductions Depress Stock Prices? An Empirical Study of Diminishing Short-Sale Constraints. *Journal of Financial and Quantitative Analysis* 36, 451–484.
- D’Avolio, G. (2002). The Market for Borrowing Stock. *Journal of Financial Economics* 66, 271–306.
- Diamond, D. W. and R. E. Verrecchia (1987). Constraints on Short-Selling and Asset Price Adjustment to Private Information. *Journal of Financial Economics* 18, 277–311.
- Easley, D., M. O’Hara, and P. Srinivas (1998). Option Volume and Stock Prices: Evidence on Where Informed Traders Trade. *Journal of Finance* 53, 431–465.
- Gagnon, L. and J. Witmer (2009). Short Changed? The Market’s Reaction to the Short Sale Ban of 2008. Working Paper, Queen’s University.

- Geczy, C. C., D. K. Musto, and A. V. Reed (2002). Stocks are Special too: an Analysis of the Equity Lending Market. *Journal of Financial Economics* 66, 241–269.
- Grundy, B. D., B. Lim, and P. Verwijmeren (2012). Do option markets undo restrictions on short sales? Evidence from the 2008 short-sale ban. *Journal of Financial Economics* 106(2), 331 – 348.
- Han, B., A. Subrahmanyam, and Y. Zhou (2017). Term Structure of Credit Default Swap Spreads and Cross Section of Stock Returns. *Journal of Financial Economics* 124, 147–171.
- Harris, L., E. Namvar, and B. Phillips (2009). Price Inflation and Wealth Transfer During the 2008 SEC Short-Sale Ban. Working Paper, University of Southern California.
- Hu, J. (2014). Does option trading convey stock price information? *Journal of Financial Economics* 111(3), 625 – 645.
- Jones, C. M. and O. A. Lamont (2002). Short-sale Constraints and Stock Returns. *Journal of Financial Economics* 66, 207–239.
- Kolasinski, A., A. Reed, and J. Thornock (2013). Can Short Restrictions Result in More Informed Short Selling? Evidence from the 2008 Regulations. *Financial Management* 42, 155–181.
- Lakonishok, J., I. Lee, N. Pearson, and A. Poteshman (2006). Option market activity. *The Review of Financial Studies* 20, 813–857.
- Lin, T.-C. and X. Lu (2016). How Do Equity Lending Costs Affect Put Options Trading? Evidence from Separating Hedging and Speculative Shorting Demands. *Review of Finance* 20, 1911–1943.
- Longstaff, F. A., S. Mithal, and E. Neis (2005). Corporate Yield Spreads: Default Risk or Liquidity? *Journal of Finance* 60, 2213–2253.
- Mayhew, S. and V. Mihov (2005). Short Sale Constraints, Overvaluation, and the Introduction of Options. Working Paper, SEC and Texas Christian University.
- Miller, E. (1977). Risk, Uncertainty, and Divergence of Opinion. *Journal of Finance* 32, 1151–1168.
- Newey, W. K. and K. D. West (1987). A Simple Positive Semi-definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix Estimator. *Econometrica* 55, 703–708.
- Norden, L. and M. Weber (2009). The Co-Movement of credit default swap, bond and stock markets: an empirical analysis. *European Financial Management* 15, 529–562.
- Ofek, E., M. Richardson, and R. F. Whitelaw (2004). Limited Arbitrage and Short Sales Restrictions: Evidence from the Options Markets. *Journal of Financial Economics* 74, 305–342.
- Pan, J. and A. Poteshman (2006). The Information in Option Volume for Future Stock Prices. *Review of Financial Studies* 19, 871–908.

- Saffi, P. A. C. and K. Sigurdsson (2011). Price Efficiency and Short Selling. *Review of Financial Studies* 24, 821–852.
- Sorescu, S. (2000). The Effect of Options on Stock Prices: 1973 to 1995. *Journal of Finance* 55, 487–514.
- Tang, D. and H. Yan (2017). Understanding Transactions Prices in the Credit Default Swaps Market. *Journal of Financial Markets* 32, 1–27.
- Xing, Y., X. Zhang, and R. Zhao (2010, 6). What Does the Individual Option Volatility Smirk Tell Us About Future Equity Returns? *Journal of Financial and Quantitative Analysis* 45, 641–662.